

Title

Protocol for a cluster randomized trial of digital health support for obesity prevention in early childhood settings in urban China

Abstract

Background: Early childhood represents a critical developmental window for obesity prevention. Schools and family environments shape dietary and activity patterns but lack effective evidence-based tools. We describe a protocol for a cluster-randomized trial of a multi-level digital health intervention targeting preschool-aged children. **Methods:** Fourteen kindergartens in an urban Chinese region will be randomly assigned to intervention (7 sites) or control (7 sites). Approximately 980 children aged 4-6 years will be enrolled from middle and senior kindergarten classrooms. The intervention comprises a cloud-based health platform providing real-time activity tracking, personalized dietary recommendations, and structured family education sessions. Implementation spans 9 months, with extended 12-month post-intervention surveillance. Data collection occurs at baseline, 4 months, 9 months, and 21 months. **Primary outcome:** between-group change in BMI z-score at the 9-month endpoint. **Secondary outcomes:** anthropometric measures, dietary quality, assessed activity levels, cost-effectiveness. Fidelity and engagement metrics will be tracked throughout. Process evaluation will identify implementation barriers and facilitators.